FASSP C++ (DIRSIG) on CIS Linux Machines

15th October 2026

1. The C++ implementation of FASSP on the CIS Linux machines is limited to the reflective solar spectral range with an implementation of equations in Kerekes and Baum, TGRS 2002.

2. A run of FASSP is controlled by a JSON configuration file (e.g., `HY_CEM_simple.json`).

3. Many *.json, *.sen, *.ref files in `/home/jpkpci/newfassp/FillFraction/`.

4. Tested on `chaffee.cis.rit.edu`, but should run on other machines. Log in using X windows terminal app.
   (from Mac Terminal app "`ssh -Y username@chaffee.cis.rit.edu`")

5. Best run under bash shell. `/usr/bin/bash`

6. Set up path: `$ export FASSP_HOME=/dirs/pkg/fassp`

7. To run from command line: `$ /dirs/pkg/fassp/bin file.json`

   OR

   To run using GUI: `$ /dirs/pkg/fassp/bin/fassp_editor` and select input `.json` file using "File" tab; start execution using "Create new ROC plot" under the "Run" tab.

   Command line prints out ROC curve data; GUI produces ROC curve plot window. GUI also allows to overlay multiple ROC curves from repeated runs using "Add to existing ROC plot" under "Run" tab.

8. Execution creates (or overwrites) `modtran_runs0` directory and several diagnostic files; recent enhancement combines data from diagnostic files into `diag_data.json` file.

9. `$ fassp_data_viewer` command hard coded to read `diag_data.json` file providing GUI interface to plot reflectance/radiance data and CEM data values.

Notes from Scott (sdbpci):

`fassp_editor` is the GUI
`fassp_cli`, which is the command-line program (and the GUI calls the program)

There are also utility programs:

- `fassp_data_viewer` is the GUI tool to plot the diagnostic (debug) data from a run
- you must run FASSP with this option enabled (e.g., `fassp_cli –enable_diagnostic_data_filefoo.json`)
- `sta_convert` is the command-line tool to convert ENVI statistics files into FASSP material files (mean and covariance)

Result from HY_CEM_simple.json
19 August 2026

